

34. Morphological operations based on OpenCV using Top hat technique.

AIM:

Morphological operations using Top hat technique.

PROGRAM:

```
import cv2

# Create a 3x3 rectangular structuring element filterSize = (3, 3)
kernel = cv2.getStructuringElement(cv2.MORPH_RECT, filterSize)

# Read the image in grayscale input_image = cv2.imread(
r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png",
cv2.IMREAD_GRAYSCALE
)

if input_image is None:
    print("Error: Image not found!") else:
    # Apply Top Hat transformation    tophat_img =
cv2.morphologyEx(input_image, cv2.MORPH_TOPHAT, kernel)

    # Save the output image    cv2.imwrite(
r"C:\Users\Susmitha\Downloads\ITA0510 Lab\tophat.jpg",
tophat_img
)

    # Display the images
cv2.imshow("Original Image", input_image)
cv2.imshow("Top Hat Image", tophat_img)

    cv2.waitKey(0)
cv2.destroyAllWindows() INPUT:
```



OUTPUT:



RESULT:

Thus the program is executed successfully.